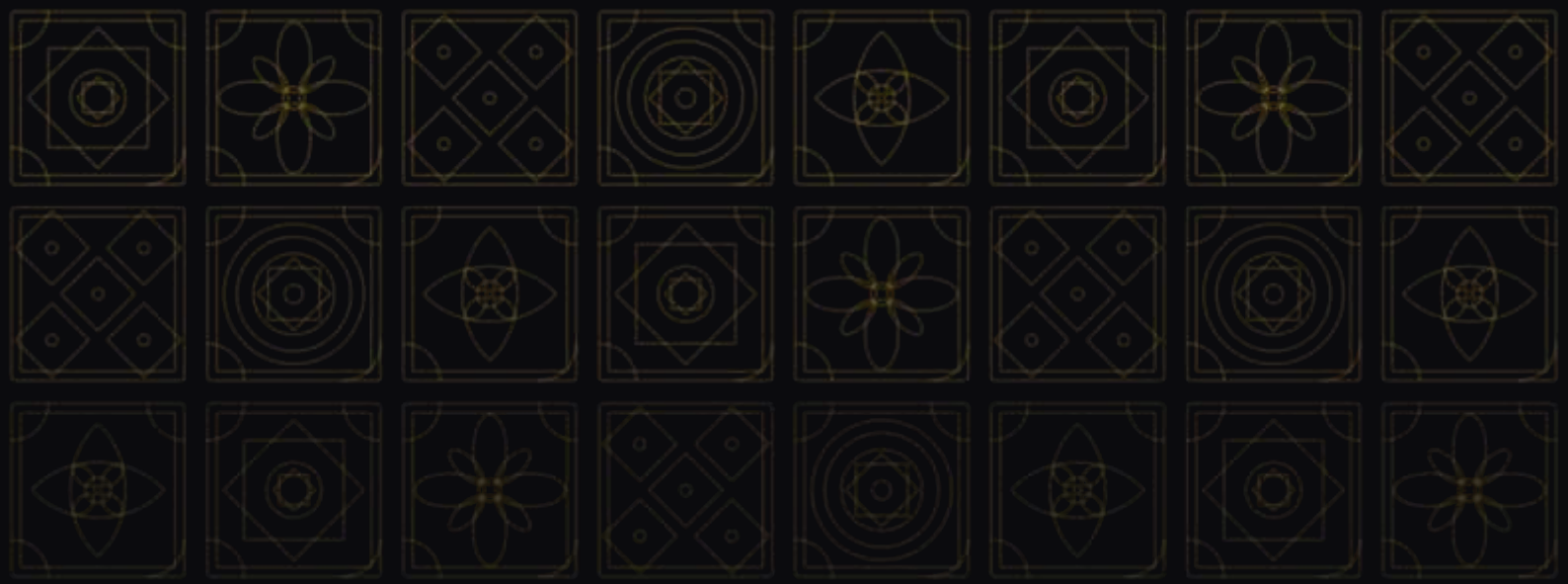




TRAINING PROGRAMME

# Zero Trust & Cloud Security

Plan a zero trust adoption and protect data and AI systems in the cloud

---

**15** HOURS

**FOR** Security managers, architects and IT leaders planning a security strategy



**Mohamed Aziz BEN HAHHA**

[mohamedazizbenhaha.netlify.app](https://mohamedazizbenhaha.netlify.app)

## Overview

This workshop helps security and IT leaders plan a move to zero trust and protect data in the cloud. Participants learn the principles of NIST SP 800-207 and the CISA Zero Trust Maturity Model, with identity at the centre, then work on cloud data security: encryption and key management, data loss prevention and data residency. They see how AI changes detection and response and how to secure AI systems themselves. The workshop closes with a final project in which each team builds a zero trust adoption roadmap for a given organisation.

## At a glance

| #     | PHASE                    | FOCUS                                                                           | HOURS |
|-------|--------------------------|---------------------------------------------------------------------------------|-------|
| 01    | Zero Trust               | NIST SP 800-207, CISA Zero Trust Maturity Model, identity-centric security      | 3 h   |
| 02    | Cloud Data & AI Security | Encryption and keys, DLP, data residency, AI for detection, securing AI systems | 6 h   |
| 03    | Final Project            | A zero trust adoption roadmap, presented to the group                           | 3 h   |
|       | Buffer                   | Catch-up, revision, extra lab time                                              | 3 h   |
| Total |                          |                                                                                 | 15 h  |

## Key facts

|                 |                                                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AUDIENCE        | Security managers, architects and IT leaders planning a security strategy                                                                                             |
| PREREQUISITES   | None. A general IT or security background is expected (course: Cybersecurity for IT Teams, recommended)                                                               |
| DURATION        | 15 h: 12 h of chapters + 3 h buffer                                                                                                                                   |
| FORMAT          | Workshop: short lectures, each followed by group exercises on an organisation case                                                                                    |
| LAB ENVIRONMENT | One laptop per participant with a browser; maturity assessment templates; trainer demonstrations in the AWS console (KMS, Macie); no account or cost for participants |
| ASSESSMENT      | An exercise validated at the end of each phase and a final roadmap presented to the group; certificate of completion for participants who complete both               |
| OUTCOME         | Participants can assess an organisation against the CISA Zero Trust Maturity Model and build a prioritised roadmap covering identity, data and AI systems             |

## PHASE 01

3 H

# Zero Trust

Participants learn what zero trust means in practice: the tenets and architecture of NIST SP 800-207, the five pillars, three cross-cutting capabilities and four maturity stages of CISA ZTMM v2.0, and why identity becomes the main control. Validated exercise: the case organisation rated on the five CISA pillars.

| #   | CHAPTER               | CONTENT                                                                    | FORMAT             | HOURS |
|-----|-----------------------|----------------------------------------------------------------------------|--------------------|-------|
| 1.1 | Zero Trust Principles | NIST SP 800-207, CISA Zero Trust Maturity Model, identity-centric security | Lecture + Workshop | 3 h   |

## PHASE 02

6 H

## Cloud Data & AI Security

Participants decide how cloud data is encrypted and who holds the keys, where data loss prevention fits and how residency rules constrain the design, then look at AI on both sides: as a help for detection and response, and as a new system to secure. Validated exercise: a data protection plan for the case organisation's most sensitive data set.

| #   | CHAPTER             | CONTENT                                                | FORMAT             | HOURS |
|-----|---------------------|--------------------------------------------------------|--------------------|-------|
| 2.1 | Cloud Data Security | Encryption, key management, DLP, data residency        | Lecture + Workshop | 3 h   |
| 2.2 | AI in Security      | AI-enabled detection and response, securing AI systems | Lecture + Workshop | 3 h   |

## PHASE 03

3 H

## Final Project

Teams of two or three build a zero trust adoption roadmap for a given organisation, reusing the workshop exercises, and present it as they would to an executive committee.

| #   | CHAPTER          | CONTENT                                                                | FORMAT  | HOURS |
|-----|------------------|------------------------------------------------------------------------|---------|-------|
| 3.1 | Assess & Plan    | Current maturity, target state, prioritised initiatives and milestones | Project | 1.5 h |
| 3.2 | Present & Review | Present the roadmap; questions and feedback                            | Project | 1.5 h |

## FINAL PROJECT

### Zero Trust Adoption Roadmap

A zero trust roadmap for a given organisation: current maturity on the CISA pillars, target state, prioritised initiatives for identity, data and AI systems, and milestones.

**STACK** NIST SP 800-207, CISA Zero Trust Maturity Model, assessment templates

**DELIVERABLE** Maturity assessment, roadmap, presentation